DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular\Supplementary Winter Examination – 2024

Course: B.Tech **Branch : Computer Engineering/Computer Science & Engg**

Semester : VII

Subject Code & Name: BTCOE705B , Deep Learning

Max Marks: 60 **Date: 21/02/2025** **Duration: 3 Hr.**

Instructions to the Students:

1. *Each question carries 12 marks.*
2. *Question No. 1 will be compulsory and include objective-type questions.*
3. *Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.*
4. *The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.*
5. *Use of non-programmable scientific calculators is allowed.*
6. *Assume suitable data wherever necessary and mention it clearly.*

					(Level/CO)	Marks
Q. 1	**Objective type questions. (Compulsory Question)**					**12**
1	What is the purpose of an activation function in a neural network?				Understand	1
	a. introduces non-linearity to the network	b. determines the output of a neuron	c. helps in back propagation	d. All of these		
2	What is the role of gradient descent in deep learning?				Analyze	1
	a. To minimize the loss function	b. To find the optimal weights for the neural network	c. To update the network's parameters	d. All of these		
3	What is the purpose of dropout regularization in deep learning?				Analyze	1
	a. To reduce overfitting	b. To increase the model's capacity	c. To improve the training speed	d. To handle imbalanced datasets		
4	Which activation function is commonly used in the hidden layers of a deep neural network?				Design	1
	a. Rectified Linear Unit	b. Sigmoid	c. Tanh	d. Softmax		
5	Which of the following is a common optimization algorithm used in deep learning?				Understand	1
	a. Gradient Descent	b. Stochastic Gradient Descent	c. Adam	d. All of these		
6	What is the purpose of a recurrent neural network (RNN)?				Understand	1
	a. To handle sequential data	b. To classify images	c. To perform dimensionality reduction	d. To generate synthetic data		
7	Which deep learning technique is used for generating new, realistic data				Understand	1

	samples?					
	a. Generative Adversarial Networks	b. Convolutional Neural Networks	c. Reinforcement Learning	d. Transfer Learning		
8	Which deep learning technique is used for unsupervised feature learning?				Analyze	1
	a.Autoencoders	b. Convolutional Neural Networks (CNNs)	c. Recurrent Neural Networks (RNNs)	d. Reinforcement Learning		
9	Which of the following is a common metric used to evaluate the performance of a classification model in deep learning?				Understand	1
	a. Accuracy	b. Mean Absolute Error (MAE)	c. Mean Squared Error (MSE)	d. R-squared		
10	CNN is mostly used when there is ?				Understand	1
	a. structured data	b. unstructured data	c. Both A and B	d. None of these		
11	Which of the following is/are Limitations of deep learning?				Understand	1
	a. Data labeling	b. Obtain huge training datasets	c. Both A and B	d. None of the above		
12	The number of nodes in the input layer is 10 and the hidden layer is 5. The maximum number of connections from the input layer to the hidden layer are				Understand	1
	a. 50	b. less than 50	c. more than 50	d. It is an arbitrary value		
Q. 2	**Solve the following.**					**12**
A)	Explain the working of deep learning model.				Understand	**6**
B)	Distinguish between Deep Learning and Machine Learning.				Analyze	**6**
Q.3	**Solve the following.**					**12**
A)	What is neuron? Explain structure of biological neuron.				Remember	6
B)	Explain architecture of feed forward neural network with necessary convention.				Understand	**6**
Q. 4	**Solve Any Two of the following.**					**12**
A)	What is back propagation learning ?				Remember	**6**
B)	How weights are updated at output layer for multilayer neural network ?				Apply	**6**
C)	Explain the concept of layer by layer pretraining mechanism.				Understand	**6**
						12

Q.5	**Solve Any Two of the following.**		
A)	Explain different variants of auto encoder.	Understand	**6**
B)	Explain architecture of Convolutional Neural Network .	Remember	**6**
C)	What are the advantages of Convolutional Neural Network over multilayer perceptron?	Understand	**6**
Q. 6	**Solve Any Two of the following.**		**12**
A)	Explain recurrent neural network.	Understand	**6**
B)	What is overfitting and underfitting problem?How is it resolved in neural networks?	Remember	**6**
C)	What is the need of normalization? Explain different kinds of normalization techniques in neural networks.	Understand	**6**
	***** End *****		